KESTREL: A Self-Flying AI Drone for Search and Rescue Without Internet

Team KESTREL

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Source code

1. Motivation

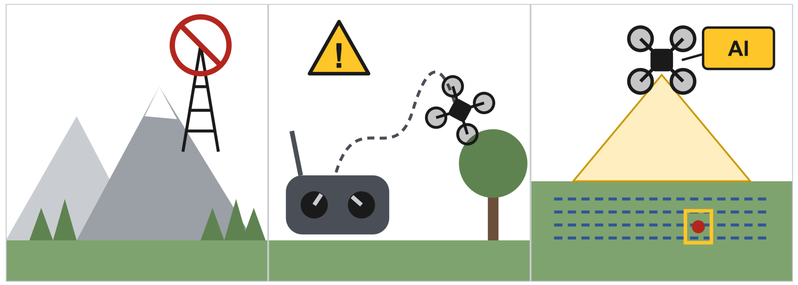


Figure 1. Steering a drone through a search is difficult even for skilled pilots, so we built KESTREL to fly the search itself (illustration).

2. Two Drone Models

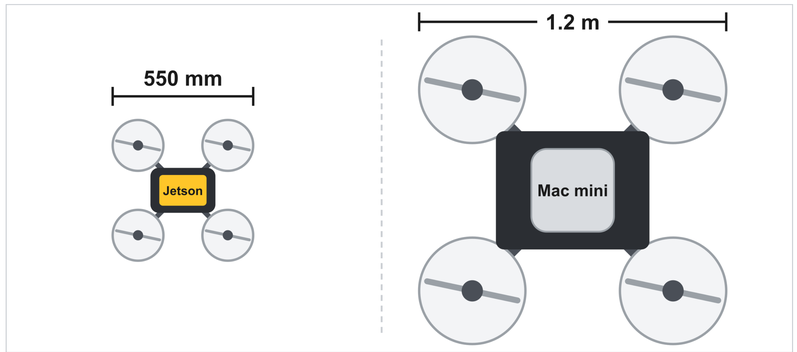


Figure 2. The two KESTREL drones, drawn to scale (top view). Same search workflow; the larger drone carries more computing power for bigger AI models.

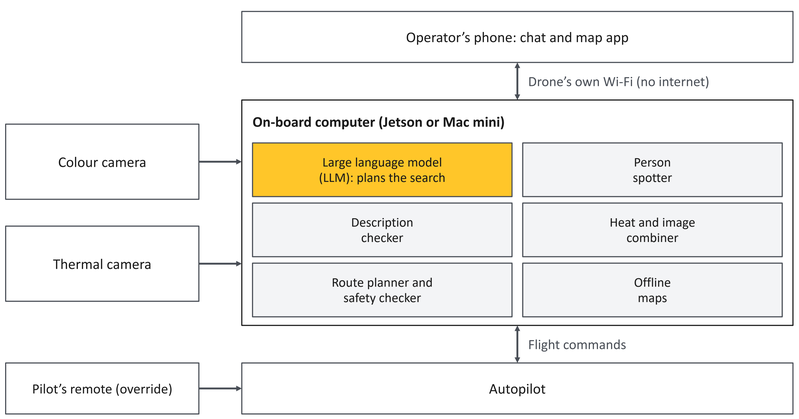


Figure 3. How the parts connect. All the AI runs on the drone itself.

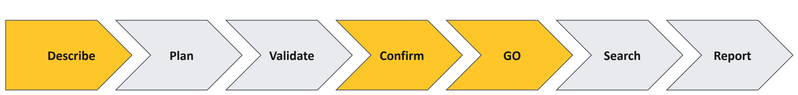


Figure 4. Operational workflow. Gold steps require a human; grey steps are automated.

3. How It Works

3.1 Planning a search by chatting

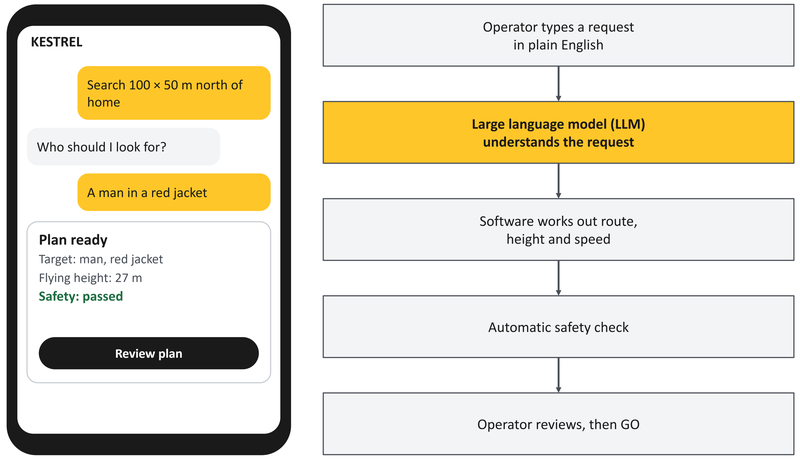


Figure 5. The large language model only understands the words. Regular software calculates every number, so the LLM never steers the drone.

3.2 Search pattern and flying height

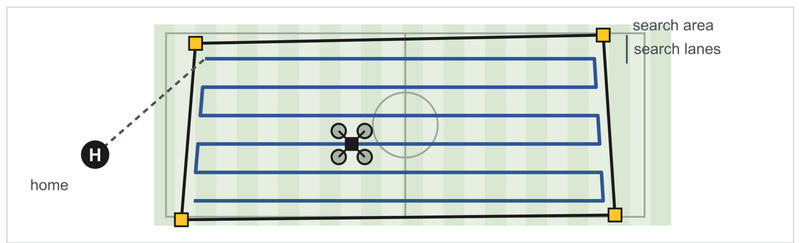
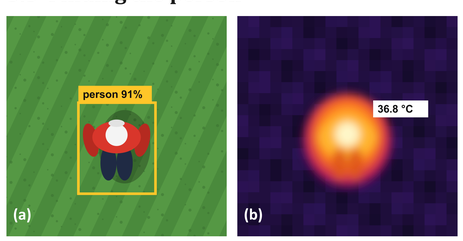


Figure 6. Back-and-forth lanes cover the whole area; every spot is photographed at least 3 times.



Figure 7. Flying height is chosen automatically: higher to spot a person, lower to see clothing or fine detail.

3.3 Finding the person



1 Camera spots a person
2 Heat confirms a warm body
3 AI compares with the description

Match: photo and map pin sent to the phone

Figure 8. (a) Colour camera and (b) thermal camera view of the same person (illustration). Both are combined and checked against the description.

4. Safety and FAA Rules

- ✓ GPS position locked
- ✓ Virtual fence set
- ✓ Airspace approval limit
- ✓ Autopilot self-test
- ✓ Cameras and AI ready
- ✓ Battery level
- ✓ Height limit: 400 ft
- ✓ Pilot's remote connected
- ✓ Route uploaded correctly
- ✓ Legal checklist

Figure 9. Checked automatically before every flight; any failure stops the launch.

FAA Part 107. A licensed drone pilot can take over at any time; the person running the search needs no flying skills.

5. Results

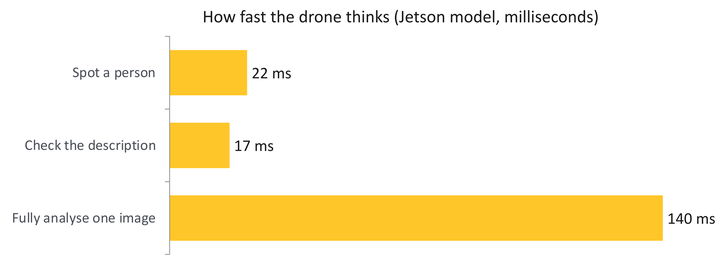


Figure 10. Each image is fully analysed in about one seventh of a second.

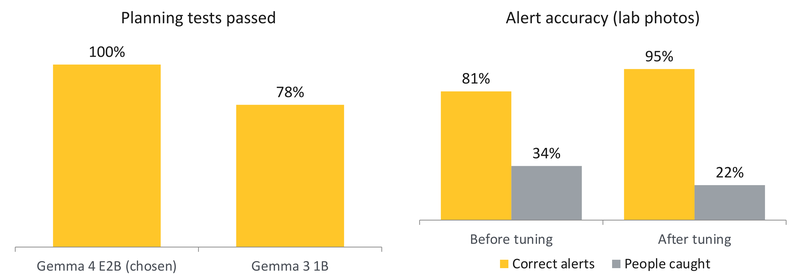


Figure 11. Left: Gemma 4 E2B passed all 16 planning tests in both runs; Gemma 3 1B passed 12 and 13. Right: tuning cut false alarms, at the cost of flagging fewer matches.

631	0	8 GB
automatic software tests	internet needed	memory (Jetson model)

6. Conclusions

- The whole search runs on a small computer on the drone, with no internet.
- The AI understands requests; proven software makes every safety decision.
- Next: tests with real aerial photos and supervised flights at KSU Marietta.

References. FAA 14 CFR Part 107 (Small Unmanned Aircraft Systems); open-source AI and autopilot software.